

Energy Use in AI

Posted by u/Bluebird8683 • 0 points • 18 comments

Hi! I'm currently working on writing a paper on energy use in AI and how it changes based on how far long in the process the AI is. Does anyone have some good sources that talk about it or have data that I can use for this?

Thank you so much for your help!

Comments

u/mobileJay77 • 1 points • 2025-11-30 20:16 UTC

Just to put it into verifiable range: I run Qwen 32B Q6 with an RTX 5090. That's about 600-700 Watt for the entire PC. It yields ~230 Tokens/s.

I don't need cooling, I live in a cold climate.

Larger models need larger VRAM and power + cooling. They would need ~4 GPUs of that kind But they should be roughly in the same order of magnitude? I would say, 10x more power could correspond to bigger hardware, but 100x more per token is where I would question efficiency. 1000x more - WTF?

u/Bluebird8683 • 1 points • 2025-11-30 22:08 UTC

thank you!

u/applitank • 1 points • 2025-11-30 13:15 UTC

For this kind of paper, your best sources will come from academic work. Try Google Scholar with terms like "AI energy consumption," "model training energy cost," "inference energy usage," and "data center energy efficiency." Set the filter to the past five years and you'll get solid, peer-reviewed papers with ready-to-use citations covering each stage of the AI lifecycle.

u/[deleted] • 1 points • 2025-11-30 11:12 UTC

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u/0LoveAnonymous0 • 1 points • 2025-11-30 11:08 UTC

Check out papers from Google's AI research team and Microsoft's sustainability reports. They've published actual numbers on training vs inference energy costs. Also look into the "Energy and Policy Considerations for Deep Learning in NLP" paper and anything from the AI Now Institute.

u/[deleted] • 1 points • 2025-11-30 09:49 UTC

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u/iswasdoes • 1 points • 2025-11-30 09:09 UTC

Data centers worldwide (not only AI ones) account for 1.5% of total energy use. It's projected to rise to 3% by 2030 because of AI.

<https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai>

u/Bluebird8683 • 1 points • 2025-11-30 22:09 UTC

thank you

u/Odd_Manufacturer2215 • 1 points • 2025-11-30 08:25 UTC

Have you seen this paper from Google on how much energy is used in models? I would take it with a pinch of salt though: [<https://www.technologyreview.com/2025/08/21/1122288/google-gemini-ai-energy/>] (<https://www.technologyreview.com/2025/08/21/1122288/google-gemini-ai-energy/>)

u/Bluebird8683 • 1 points • 2025-11-30 22:09 UTC

yeah, second agendas and bias are a thing. thank you

u/brockchancy • 1 points • 2025-11-30 04:25 UTC

Daily use / inference

1. **IEA – “Energy and AI” (2025 report)** [<https://www.iea.org/reports/energy-and-ai>] (https://www.iea.org/reports/energy-and-ai?utm_source=chatgpt.com) → *Big-picture look at how much electricity AI is using today and how fast it might grow, written by the main global energy agency governments rely on.*
2. **Columbia SIPA – Center on Global Energy Policy** **“Projecting the Electricity Demand Growth of Generative AI Large Language Models in the US”** [<https://www.energypolicy.columbia.edu/projecting-the-electricity-demand-growth-of-generative-ai-large-language-models-in-the-us/>] (https://www.energypolicy.columbia.edu/projecting-the-electricity-demand-growth-of-generative-ai-large-language-models-in-the-us/?utm_source=chatgpt.com) → *Zooms in on chatbots/LLMs specifically and estimates how much extra electricity they could add to the U.S. grid when millions of people use them every day.*
3. **EU Commission – “In focus: Data centres – an energy-hungry challenge”** [https://energy.ec.europa.eu/news/focus-data-centres-energy-hungry-challenge-2025-11-17_en] (https://energy.ec.europa.eu/news/focus-data-centres-energy-hungry-challenge-2025-11-17_en?utm_source=chatgpt.com) → *Explains why data centers (where AI lives) are so power-hungry, and how governments in Europe are starting to worry about and regulate their energy use.*

Training runs / lifecycle impact

1. **Stanford HAI – AI Index Report 2024 (full PDF, see Environmental Impact chapter)** [https://hai-production.s3.amazonaws.com/files/hai_ai-index-report-2024-smaller2.pdf](https://hai-production.s3.amazonaws.com/files/hai_ai-index-report-2024-smaller2.pdf?utm_source=chatgpt.com) → *Has a chapter that adds up how much energy and carbon it takes to* ***train*** *big AI models and compares different systems so you can see how large these one-time hits are.*
2. **Jiang et al. – “Preventing the Immense Increase in the Life-Cycle Energy Consumption of Artificial Intelligence”** [https://www.researchgate.net/publication/379912169_Preventing_the_Immense_Increase_in_the_Life-Cycle_Energy_and_Carbon_Footprints_of_LLM-Powered_Intelligent_Chatbots] (https://www.researchgate.net/publication/379912169_Preventing_the_Immense_Increase_in_the_Life-Cycle_Energy_and_Carbon_Footprints_of_LLM-Powered_Intelligent_Chatbots)

Cycle_Energy_and_Carbon_Footprints_of_LLM-Powered_Intelligent_Chatbots) → *Looks at AI from “birth to retirement” — training + years of usage — and argues how we could design AI so its total lifetime energy use doesn’t explode.*

3. **Li et al. — “Making AI Less ‘Thirsty’: Uncovering and Addressing the Secret Water Footprint of AI Models”** arXiv PDF: [<https://arxiv.org/pdf/2304.03271>](<https://arxiv.org/pdf/2304.03271>?utm_source=chatgpt.com) → *Shows that training big models doesn’t just use electricity — it also uses a lot of cooling water — and gives real estimates (like “training Model X used about as much water as...”), which makes the impact more concrete.*

u/Bluebird8683 • 2 points • 2025-11-30 22:08 UTC

thank you so much!!!

u/brockchancy • 1 points • 2025-11-30 22:41 UTC

Imo this gives all the data we need to push for Hybrid/Dry cooling systems and A National Grid update. Problems exist but all of them are engineerable problems.

u/Titanium-Marshmallow • 1 points • 2025-11-30 03:38 UTC

hmmm. did you try searching using an LLM?

u/HoldTheMayo25 • 3 points • 2025-11-30 03:26 UTC

Focus on the distinction between Training (a massive, one-time energy cost) and Inference (the ongoing cost of answering user queries). While training a large model like GPT-3 consumed about 1,287 MWh, industry estimates suggest that inference actually accounts for 80-90% of a model's lifecycle energy use due to the sheer volume of daily users.

For citations, look up Sasha Luccioni's work (Hugging Face) for data on inference and task-specific costs (e.g., how image generation uses far more power than text), and Patterson et al. (2021) for the foundational benchmarks on training emissions.

* Patterson et al. (2021): Best for data on Training energy.

* Luccioni et al. (2023/2024): Best for data on Inference and Task Comparisons (Text vs. Image).

* Strubell et al. (2019): The seminal paper that started the conversation on AI energy use (good for historical context).

u/mobileJay77 • 1 points • 2025-11-30 18:56 UTC

Is that the Patterson who wrote books on computer architecture?

u/Bluebird8683 • 1 points • 2025-11-30 03:51 UTC

thank you!

u/AutoModerator • 1 points • 2025-11-30 03:10 UTC

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find it useful?

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